

ISYE-6767 QCF Interim Project 1
Implementing a Dynamic Delta Hedging Strategy

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Problem Addressed:

This project focuses on implementing a dynamic delta-hedging strategy for European call options. The objective is to replicate the option payoff by continuously adjusting the position in the underlying asset, thereby mitigating market risk. The work involves two stages: first, simulating stock price dynamics under the Black-Scholes framework and validating the hedge in a theoretical setting; second, applying the hedge to real market data (GOOG, 2011) to evaluate practical performance and compute hedging P&L and cumulative hedging errors.

Model Used:

- **Black–Scholes–Merton Model:** Used for option pricing and delta computation.

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

- **Discrete-time stochastic price simulation:** Stock paths are simulated using the additive form specified in the project instruction:

$$S_{t+\Delta t} = S_t + \mu S_t \Delta t + \sigma S_t \sqrt{\Delta t} Z_t$$

This allows testing the hedging strategy under controlled, model-consistent price dynamics.

- **Binary-search:** Used to search for the implied volatility of the option.

Option_Price.h and Option_Price.cpp

1. Classes

- **Option Class**
Stores core option parameters: strike price, underlying price, risk-free rate, time to maturity, and volatility.
- **PricingMethod Interface**
Abstract interface that defines the pricing behavior for option pricing models.
- **Option_Price Class**
Inherits from Option and implements Black-Scholes-Merton pricing.
Returns both option price and delta through BSM_Pricer().

2. Key Methods

- Constructor initializes option parameters
- Black-Scholes pricer (BSM_Pricer) outputs (price, delta) pair
- Handles $T \leq 0$ case (exercise payoff & terminal delta)

Simulation.h and Simulation.cpp

1. Functions: *simulate_and_hedge()*: generates stock paths and option values under project-specified additive model, computes delta hedge path-wise, tracks bank account and cumulative hedging error, and outputs CSV files:
 - *simulated_stock_prices.csv*
 - *simulated_option_prices.csv*
 - *hedging_errors_paths.csv*
 - *hedging_errors_terminal.csv*

Data_Read.h and Data_Read.cpp

1. Data Structures
 - *TotalInterestRateData* — daily risk-free rate
 - *TotalStockData* — GOOG adjusted close price
 - *TotalOptionData* — GOOG option quotes (bid/ask, strike, expiration)
2. Functions
 - *read_interest_data()* loads *interest.csv*
 - *read_stock_data()* loads *sec_GOOG.csv*
 - *read_option_data()* loads *op_GOOG.csv*

utils.h and utils.cpp

1. Functions
 - *normCdf()* — Standard normal CDF
 - *getPreviousDateIndex()* — finds nearest trading date index
 - *dateCount()* — computes business-day difference
 - *optionMidQuote()* — returns option mid-quote for given date/strike
 - *volBinarySearchRobust()* — implied volatility solver using robust expanding binary search (handles NA cases)

PNL_Cal.h and PNL_Cal.cpp

1. Main Function: *deltaHedgingPortfolioConstruct()*
2. Inputs
 - $t_0, t_{\square}$: Start and end trading dates for the hedging period
 - *exp_date*: Option expiration date
 - *flag*: Option type indicator ("C" for call)
 - *strike_price*: Strike price for the contract
 - *rates* — interest rate data
 - *stocks* — stock price data

- options — option quote data
3. Operations
 - Aligns trading dates
 - Computes implied vol daily via binary search
 - Computes delta each day
 - Tracks bank balance & hedging error
 - Writes result.csv

unit_test.h

1. Functions
 - *testFun()* — checks accuracy of implied vol and delta vs known values
 - *testCases()* — runs multiple sanity checks

Unit tests confirm IV and delta match expected values within tolerance.

main.cpp

- Runs unit tests - *testCases()*;
- Runs simulation engine - *simulate_and_hedge(simulation_num, S0, T, mu, sg, r, N, K)*;
- Loads market data:
 - *read_interest_data(rates)*;
 - *read_stock_data(stocks)*;
 - *read_option_data(options)*;
- Executes real data hedging: *deltaHedgingPortfolioConstruct("2011-07-05", "2011-07-29", "2011-09-17", "C", 500.0, rates, stocks, options)*;
- Outputs all result CSVs - result.csv

Results

1. Task 1 — Simulation

- Simulated stock price paths (100 out of 1000 shown)

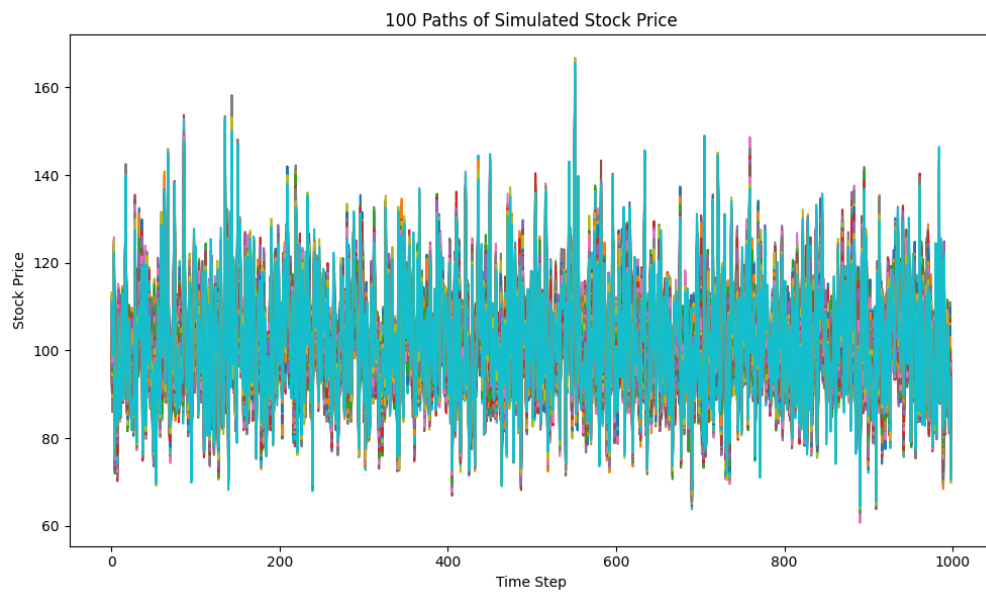


Figure 1. Simulated stock price paths under the Black-Scholes model

- Simulated option price paths

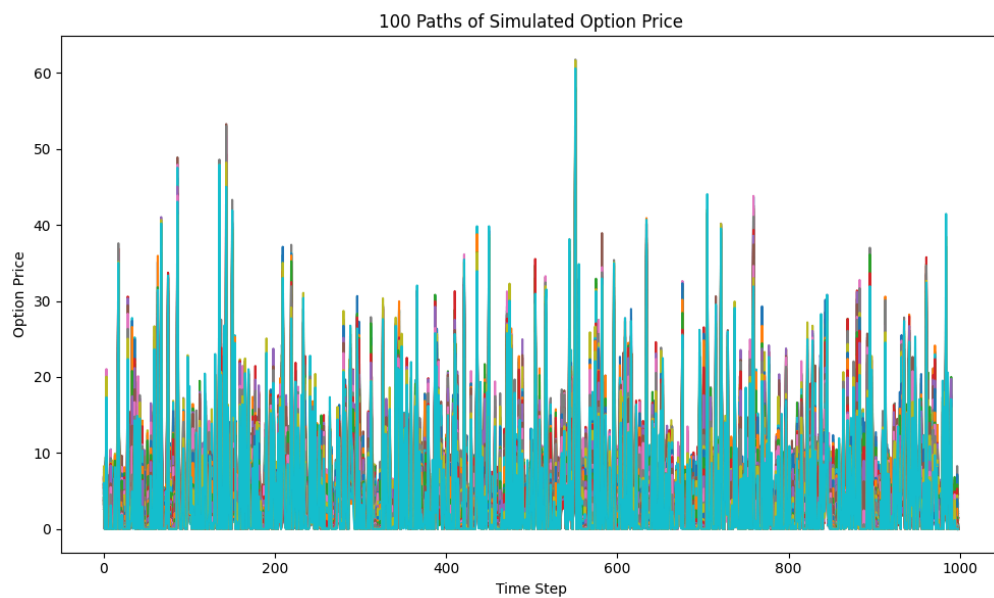


Figure 2. Simulated option price paths corresponding to the simulated stock paths

These two figures demonstrate the expected option price volatility behavior under simulation.

- Hedging error distribution at maturity centered near zero

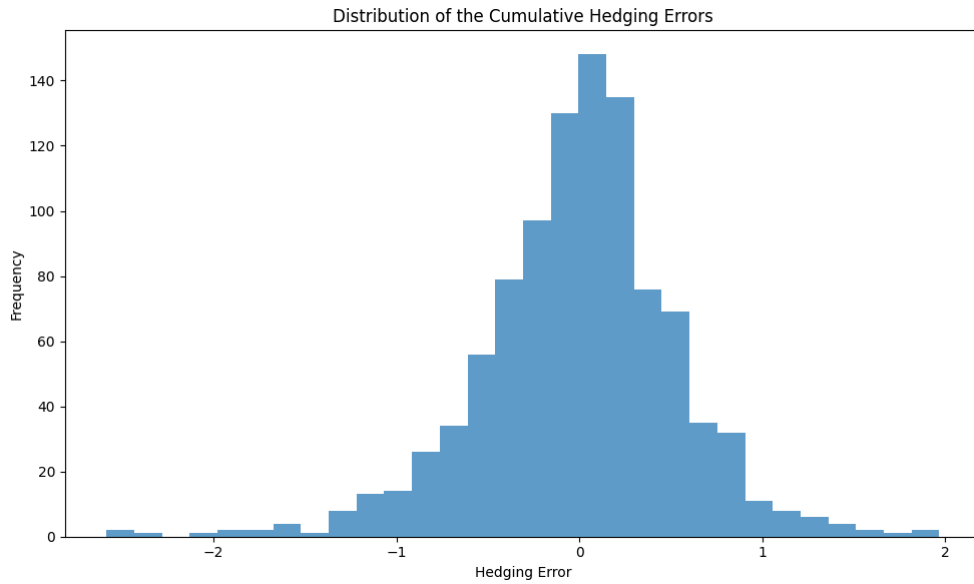


Figure 3. Distribution of cumulative hedging errors across simulated paths

This plot supports the theoretical claim: under BSM, continuous-time hedging approximated in discrete time yields approximately zero-mean hedging error.

- Task 2 — Real Market Data (GOOG 2011)
 - Daily implied volatility curve

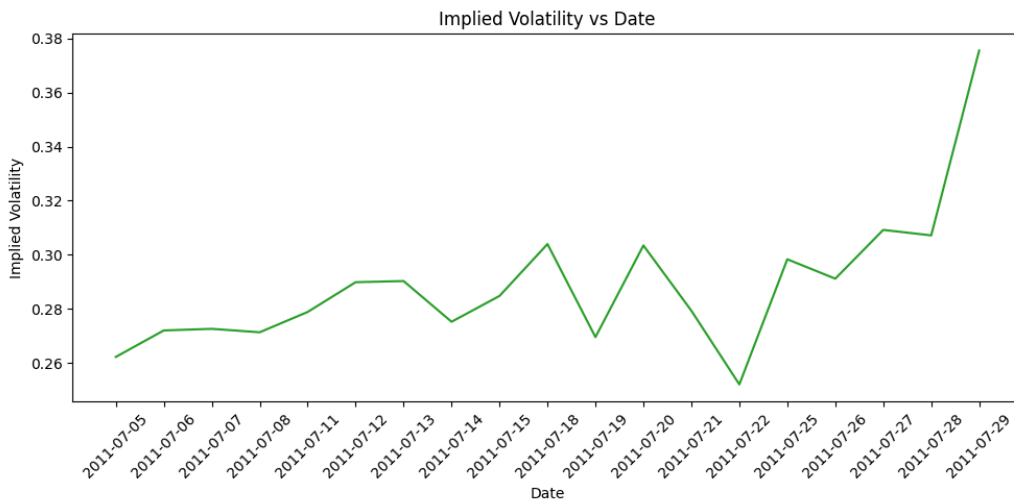


Figure 4. Daily Implied Volatility

Shows market volatility is not constant, illustrating deviation from the Black-Scholes assumption.

- Daily delta curve

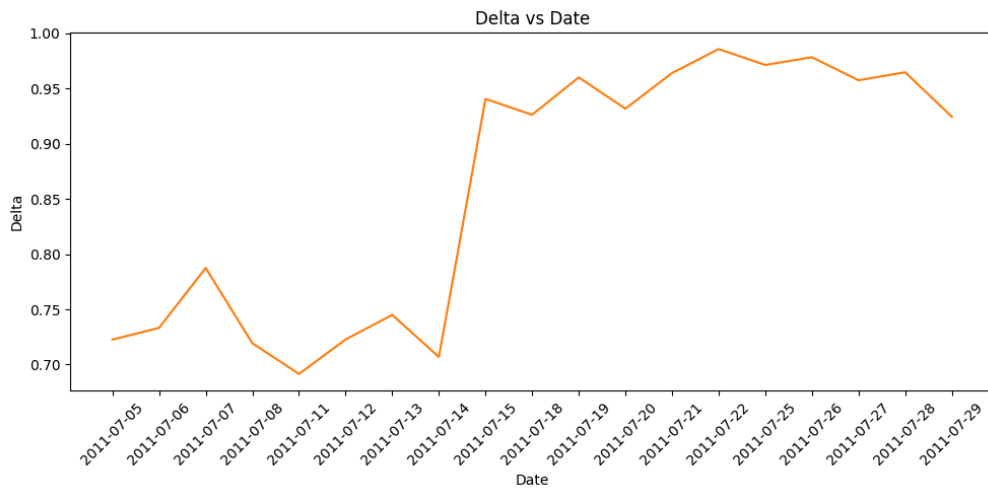


Figure 5. Daily Delta

Delta increases as the option becomes more in-the-money and time to maturity decreases, reflecting greater sensitivity to underlying price movements.

- Hedged PNL shows smoother return vs unhedged short call

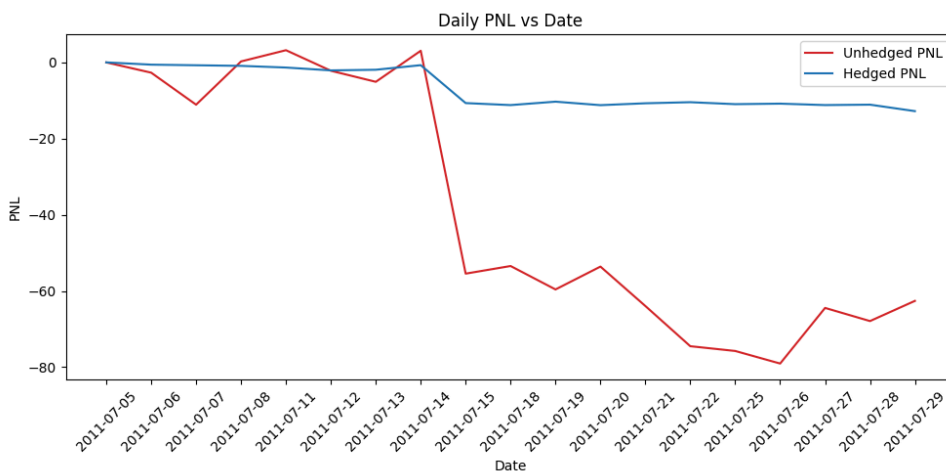


Figure 6. Daily PNL: Hedged vs. Unhedged

The unhedged short call incurs large losses during price increases.

The hedged portfolio remains relatively stable, confirming that delta-hedging reduces risk in practice.

date	S	V	ImpliedVol	delta	HE	PNL	PNL_with_hedge
7/5/11	532.44	44.2	0.262203	0.722613	0	0	0
7/6/11	535.36	46.9	0.271998	0.733277	-0.592537	-2.7	-0.592537
7/7/11	546.6	55.3	0.272588	0.787526	-0.75311	-11.1	-0.75311
7/8/11	531.99	43.95	0.271319	0.71936	-0.911853	0.25	-0.911853
7/11/11	527.28	41	0.278837	0.691477	-1.35233	3.2	-1.35233
7/12/11	534.01	46.4	0.289854	0.722731	-2.10088	-2.2	-2.10088
7/13/11	538.26	49.3	0.290272	0.745022	-1.93171	-5.1	-1.93171
7/14/11	528.94	41.15	0.275212	0.706872	-0.727562	3.05	-0.727562
7/15/11	597.62	99.65	0.284765	0.940683	-10.6816	-55.45	-10.6816
7/18/11	594.94	97.65	0.303965	0.926389	-11.2055	-53.45	-11.2055
7/19/11	602.55	103.8	0.26953	0.960205	-10.3084	-59.6	-10.3084
7/20/11	595.35	97.8	0.30345	0.931852	-11.2251	-53.6	-11.2251
7/21/11	606.99	108.15	0.27929	0.964157	-10.7319	-63.95	-10.7319
7/22/11	618.23	118.7	0.252031	0.985862	-10.4487	-74.5	-10.4487
7/25/11	618.98	119.95	0.298333	0.971489	-10.9632	-75.75	-10.9632
7/26/11	622.52	123.25	0.291152	0.978467	-10.8281	-79.05	-10.8281
7/27/11	607.22	108.65	0.3092	0.957595	-11.2028	-64.45	-11.2028
7/28/11	610.94	112.1	0.307171	0.964875	-11.0945	-67.9	-11.0945
7/29/11	603.69	106.8	0.375561	0.924645	-12.794	-62.6	-12.794

Table 1. Sample output from delta-hedging on GOOG (result.csv excerpt)

The hedged portfolio remains near zero throughout the trading period, whereas the unhedged short call incurs significant losses as the stock price rises. This demonstrates the effectiveness of dynamic delta hedging in practice.

```
Running Task 2 unit tests...
Running test cases for Task 2...
-----
Option type: C
Searched implied volatility: 0.200000 | Target: 0.200000
Calculated delta: 0.946539 | Target: 0.946539
✓ Test passed!
-----
Option type: P
Searched implied volatility: 0.250000 | Target: 0.250000
Calculated delta: -0.046540 | Target: -0.046540
✓ Test passed!
=====
✓ All test cases passed!

Task 1 simulation...
Generated: simulated_stock_prices.csv, simulated_option_prices.csv,
(.venv) (base) sunyijun@Yijuns-MacBook-Air system_design %
```

Figure 7 Output

Conclusion

This project implemented a dynamic delta-hedging strategy based on the Black-Scholes-Merton model and tested it both in simulation and using real market data. The simulation results confirmed theoretical expectations: hedging errors are centered around zero when model assumptions hold.

In the empirical test with GOOG data, the hedge substantially reduced PNL volatility and downside risk, despite market frictions such as changing implied volatility and discrete rebalancing. These results highlight both the strengths of delta hedging as a risk-management tool and the practical deviations from Black-Scholes assumptions in live market conditions.